

An Explicit Formula for the Loday Assembly

Virgil Chan

✧ Introduction

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unital, associative

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The space K_R^+ is actually an infinite loop space

$$K_R^+ \cong \Omega BGL(\Sigma R)^+ \quad (\text{Gersten, Wagoner 1972})$$

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suspension of R

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So we get an Ω -spectrum :

$$(K_R^{GW})_n := K_{\Sigma^n R}^+ \quad (\text{Gersten-Wagoner non-connective K-theory spectrum})$$

with $\pi_i(K_R^{GW}) \cong K_i(R)$.

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Homotopy groups of LHS can be computed by the Atiyah-Hirzebruch spectral sequence:

$$E_{p,q}^2 \cong H_p(BG; K_q(R)) \Rightarrow \pi_{p+q}(BG_+ \wedge K_R^{GW}).$$

Conjecture (K-theoretic Novikov Conjecture, Hsiang 1984)

If R is regular and G is torsion-free, then

$$\pi_i(\alpha_L) \otimes \text{id}_{\mathbb{Q}} : \pi_i(BG_+ \wedge K_R^{GW}) \otimes \mathbb{Q} \rightarrow K_i(R[G]) \otimes \mathbb{Q}$$

is injective for all i .



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Thm (Bökstedt - Hsiang - Madsen 1993)

True if G is homologically finite.



Conjecture (Classical Farrell-Jones Conjecture)

If R is regular and G is torsion-free, then

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Thm (Bartels-Lück-Reich 2008)

True if G is word-hyperbolic.



* Our Results

We contribute to these problems by giving an explicit description of $\pi_i(\alpha_L)$ when restricted onto a subgroup of $\pi_i(BG_+ \wedge K_R^{GW})$.

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$\uparrow g$	$K_5(R)$	$H_1(BG; K_5(R))$	$H_2(BG; K(R))$
	$K_4(R)$	$H_1(BG; K_4(R))$	$H_2(BG; K(R))$
	$K_3(R)$	$H_1(BG; K_3(R))$	$H_2(BG; K(R))$
	$K_2(R)$	$H_1(BG; K_2(R))$	$H_2(BG; K(R))$
	$K_1(R)$	$H_1(BG; K_1(R))$	$H_2(BG; K(R))$
	$K_0(R)$	$H_1(BG; K_0(R))$	$H_2(BG; K(R))$
	$\rightarrow p$		

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$\pi_i(\alpha_L)$ is induced by $R \hookrightarrow R[G]$.

$K_5(R)$	$G_{ab} \otimes K_5(R)$	$H_2(BG; K(R))$
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Our results tell what $\pi_i(\alpha_L)$ is doing on these.

g			
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$E_{p,g}^2 \cong H_p(BG; K_g(R))$

More precisely, the subgroup we are interested in is a quotient of $E_{1,i}^2 \cong G_{ab} \otimes K_i(R)$:

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$$E_{1,i}^\infty$$



$$K_{i+1}(R[G])$$



$$\pi_{i+1}(\alpha_L)$$

$$\pi_{i+1}(BG_+ \wedge K_R^{GW})$$

More precisely, the subgroup we are interested in is a quotient of $E_{1,i}^2 \cong G_{ab} \otimes K_i(R)$:

$$\begin{array}{ccc}
 E_{1,i}^2 \cong G_{ab} \otimes K_i(R) & \xrightarrow{\quad \overline{\Phi}_i \quad} & K_{i+1}(R[G]) \\
 \downarrow & \curvearrowright & \uparrow \pi_{i+1}(\alpha_L) \\
 E_{1,i}^\infty & \xrightarrow{\quad \quad \quad} & \pi_{i+1}(BG_+ \wedge K_R^{GW})
 \end{array}$$

We describe the filler of this commutative diagram.

Prop (The Case of Infinite Cyclic Group)

When $G = \langle t \rangle$ is the infinite cyclic group, the **filler**

$$\begin{array}{ccccc}
 G_{ab} \otimes K_i(R) & \cong & K_i(R) & \xrightarrow{\Phi_i} & K_{i+1}(R[t^{\pm 1}]) \cong K_{i+1}(R[G]) \\
 \downarrow & & & \curvearrowright & \uparrow \alpha_L \\
 E_{1,i}^{\infty} & & \xrightarrow{\quad} & & \pi_{i+1}(B\langle t \rangle_+ \wedge K_R^{GW})
 \end{array}$$

is induced by

$$(t, [f]) \mapsto [f] \star \{\mu_t\},$$

where $\mu_t \in \text{Aut}(R[t^{\pm 1}])$ is multiplication by t^{-1} , and $\star$ is the Loday product. □

The Loday $\star$ -product is defined as follows :

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$$\otimes : GL(P, R) \times GL(S, S) \longrightarrow GL(R \otimes S)$$

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$$\gamma : BGL(R)^+ \times BGL(S)^+ \longrightarrow BGL(R \otimes S)^+$$

$$\gamma(x, y) := f(x, y) - f(x_0, y) - f(x, y_0)$$

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$$\gamma : BGL(R)^+ \times BGL(S)^+ \longrightarrow BGL(R \otimes S)^+ \quad \text{base-point}$$

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$\downarrow$

$$\hat{\gamma} : BGL(R)^+ \wedge BGL(S)^+ \longrightarrow BGL(R \otimes S)^+$$

(Well-defined up to "weak homotopy")

We then extend $\hat{\gamma} : BGL(R)^+ \wedge BGL(S)^+ \rightarrow BGL(R \otimes S)^+$
by the Gersten-Wagoner Delooping $\circ$

$$K_R^+ \wedge K_S^+ \xrightarrow[\quad \uparrow \quad]{\gamma_L} K_{R \otimes S}^+$$

$\mathbb{Z} \qquad \qquad \qquad \mathbb{Z}$

$$\Omega BGL(\Sigma R)^+ \wedge \Omega BGL(\Sigma S)^+ \xrightarrow[\Omega^2 \hat{\gamma}]{\quad} \Omega^2 BGL(\Sigma^2(R \otimes S))^+$$

Then,

$$\begin{aligned} \star : K_i(R) \otimes K_j(S) &\longrightarrow K_{i+j}(R \otimes S) & i, j \geq 0 \\ [f] \otimes [g] &\longmapsto [\gamma_L \circ (f \wedge g)] \end{aligned}$$

$$\boxed{\text{e.g.}} \quad \star: K_0(R) \otimes K_0(S) \rightarrow K_0(R \otimes S)$$

$$[P] \otimes [Q] \mapsto [P \otimes Q]$$

$$\star: K_0(R) \otimes K_1(S) \rightarrow K_1(R \otimes S)$$

$$[P] \otimes \{v\} \mapsto \{P \otimes v^{-1} + (I-P) \otimes \text{id}\}$$

$$\star: K_1(R) \otimes K_1(S) \rightarrow K_2(R \otimes S)$$

$$\{u\} \otimes \{v\} \mapsto \{u, v\}_{\text{st}} \text{ (Steinberg Symbol)}$$



Thm (C)

For any group G , the filler

$$\begin{array}{ccc}
 G_{ab} \otimes K_i(R) & \xrightarrow{\quad \overline{\Phi}_i \quad} & K_{i+1}(R[G]) \\
 \downarrow & \curvearrowright & \uparrow \alpha_L \\
 E_{1,i}^\infty & \longrightarrow & \pi_{i+1}(BG_+ \wedge K_R^{GW})
 \end{array}$$

is induced by

$$(g, [f]) \mapsto [f] \star \{\mu_g\}.$$



Proof (Outline)

We always have

$$G \cong \text{Hom}(\langle t \rangle, G)$$

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We will bootstrap from the case $G = \langle t \rangle$.

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Can then form

$$\pi_{i+1}(\mathcal{S}' \wedge K_R^{GW})$$



$$\pi_{i+1}(B\langle t \rangle \wedge K_R^{GW})$$



$$K_{i+1}(R[t^{\pm 1}])$$



$$\pi_{i+1}\left(\left(\bigvee_{g \in G} \mathcal{S}'\right) \wedge K_R^{GW}\right)$$



$$\pi_{i+1}(BG \wedge K_R^{GW})$$



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We always have

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$$\begin{array}{ccc} \pi_{i+1}(\mathcal{S}' \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}\left(\left(\bigvee_{g \in G} \mathcal{S}'\right) \wedge K_R^{GW}\right) \\ \downarrow & & \downarrow \\ \pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}(BG \wedge K_R^{GW}) \\ \downarrow & & \downarrow \\ K_{i+1}(R[t^{\pm 1}]) & \xrightarrow{(\varphi_g)_*} & K_{i+1}(R[G]) \end{array}$$

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 \downarrow & \text{inclusion of 1-skeleton} \rightsquigarrow & \downarrow \\
 \pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}(BG \wedge K_R^{GW}) \\
 \alpha_L \downarrow & & \downarrow \alpha_L \\
 K_{i+1}(R[t^{\pm 1}]) & \xrightarrow{(\varphi_g)_*} & K_{i+1}(R[G])
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 \downarrow \swarrow \text{inclusion of the} & & \downarrow \\
 & \text{1-cell associated to the generator } t & \\
 \pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}(BG \wedge K_R^{GW}) \\
 \downarrow \alpha_L & & \downarrow \alpha_L \\
 K_{i+1}(R[t^{\pm 1}]) & \xrightarrow{(\varphi_g)_*} & K_{i+1}(R[G])
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$$K_i(R)$$



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$$\alpha_L \downarrow$$

$$K_{i+1}(R[t^{\pm 1}])$$

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$$\bigoplus_{g \in G} K_i(R)$$



$$\pi_{i+1}(BG \wedge K_R^{GW})$$

$$\downarrow \alpha_L$$

$$K_{i+1}(R[G])$$

Proof (Outline)

We always have

$$G \cong \text{Hom}(\langle t \rangle, G)$$

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Can then form

$$\begin{array}{ccc}
 [f] \in K_i(R) & \xrightarrow{(\varphi_g)_*} & \bigoplus_{g \in G} K_i(R) \ni [f]_g \\
 \downarrow & & \downarrow \\
 [f] \in \Pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \Pi_{i+1}(BG \wedge K_R^{GW}) \\
 \downarrow & & \downarrow \alpha_L \\
 [f] \star \{\mu_t\} \in K_{i+1}(R[t^{\pm 1}]) & \xrightarrow{(\varphi_g)_*} & K_{i+1}(R[G]) \ni [f]_g \star \{\mu_g\}
 \end{array}$$

(By Proposition)

(By Commutativity)



Corollary ("Lück-Riech 2005")

For a regular ring R , $\pi_1(\alpha_L)$ is given by

$$K_1(i) \oplus \overline{\Phi}_1 : K_1(R) \oplus [G_{ab} \otimes K_0(R)] \rightarrow K_1(R[G])$$

for which $\overline{\Phi}_1$ is induced by

$$\begin{aligned} G \times K_0(R) &\longrightarrow K_1(R[G]) \\ (g, [P]) &\longmapsto \left\{ \begin{array}{l} \eta_g : P \otimes R[G] \rightarrow P \otimes R[G] \\ x \otimes u \mapsto x \otimes ug^{-1} \end{array} \right\} \end{aligned}$$



Corollary (C, Bijectivity / Injectivity for $C_2^{\oplus n}$)

The map

$$\pi_2(\alpha_L) : \pi_2(BC_2^{\oplus n} \wedge K_{\mathbb{Z}}^{GW}) \rightarrow K_2(\mathbb{Z}[C_2^{\oplus n}])$$

is bijective when $n=1$. When $n>1$, the composition

$$K_2(\mathbb{Z}) \oplus [C_2^{\oplus n} \otimes K_1(\mathbb{Z})] \hookrightarrow \pi_2(BC_2^{\oplus n} \wedge K_{\mathbb{Z}}^{GW}) \xrightarrow{\pi_2(\alpha_L)} K_2(\mathbb{Z}[C_2^{\oplus n}])$$

is injective.



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is injective. □

Extension Problem From the AHSS :

$$0 \rightarrow K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \rightarrow \pi_2(BG_+ \wedge K_{\mathbb{Z}}^{GW}) \rightarrow H_2(BG; K_0(\mathbb{Z})) \rightarrow 0$$

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$$\Rightarrow \pi_2(BC_{2+} \wedge K_{\mathbb{Z}}^{GW}) \cong K_2(\mathbb{Z}) \oplus [C_2 \otimes K_1(\mathbb{Z})] \cong C_2 \oplus C_2$$

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$$\text{Dunwoody (1975)} : K_2(\mathbb{Z}[C_2]) \cong \langle \{-1, -1\}_{st}, \{-1, 9\}_{st} \rangle \cong C_2 \oplus C_2$$

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$$\cong \downarrow K_2(i) \oplus \Phi_2$$

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